Debadeep Chaudhury

Software Developer

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Software developer with 5 years of research programming in Python, R and Bash, now studying C and systems programming at 42 Warsaw. In 2026, team manager for 4 hackathon teams (3rd place; 1st in track and 2nd overall; top 10 in track twice), and builder of self-hosted full-stack and AI-agent apps with FastAPI, React, TypeScript, PostgreSQL and Docker.

SKILLS

Languages: Python, TypeScript, JavaScript, C, SQL, R, Bash

Backend and data: FastAPI, REST and OpenAPI, Next.js, PostgreSQL, PostGIS, Supabase (row-level security, migrations), SQLite, ETL pipelines, JWT/JWKS authentication, pytest

Frontend: React 19, Vite, Recharts, HTML, CSS

AI and automation: tool-calling agents, retrieval-augmented generation (RAG), Model Context Protocol (MCP) servers, n8n, OpenRouter, local models

Infrastructure: Docker, Docker Compose, Linux (Ubuntu), Git, CI/CD (GitHub Actions), Caddy, Cloudflare Tunnel, Tailscale, Railway, Vercel; AWS, Azure and GCP (Linux VMs)

Ways of working: Agile/Scrum, code review through pull requests, automated testing

EXPERIENCE

Independent Software Developer · 42 Warsaw — Jan 2026 – Present

- Built MTG Azor, a self-hosted tool-calling AI agent that answers Magic: The Gathering rules questions from the Comprehensive Rules (RAG), live Scryfall card data and official rulings, and verifies every citation against the sources it fetched (186 commits, TypeScript and Python)
- Served one API to two clients, a streaming web app and a Discord bot, with a switch between local and hosted language models, all packaged in Docker
- Published Delta Notes, a Pebble smartwatch app in C that sends dictated notes to an Obsidian vault through an n8n webhook; tested on real hardware and live on the Pebble Appstore
- Automated a voice-note research pipeline (Pebble ring → n8n → AI research through MCP servers → tagged Obsidian note) with 2 custom MCP server images published to GitHub Container Registry
- Built a design-system library with 17 design systems that pass WCAG AA in both themes, a web dashboard, a browser capture extension and an OAuth-protected MCP connector for AI assistants
- Run a self-hosted Ubuntu home server: 25 Docker containers in 16 Compose stacks (Gitea, n8n, MinIO, SearXNG, Home Assistant with MQTT) behind Caddy and 5 Cloudflare Tunnels, with Tailscale access, Beszel monitoring, PostgreSQL 18 and Ollama
- Coordinate a 3-person team on Warsaw in Numbers, a district comparison app (Python ETL, Supabase PostgreSQL, FastAPI), with a locked main branch and review-gated pull requests

Workshop Instructor (volunteer) · 42 Warsaw — Sept 2026

- Designed and taught the Guided Coding Workshop to 12 participants: one FastAPI + React app built three times, with a web chat, an editor agent and a command-line agent, plus a handout, slides and a free-tools setup guide

HACKATHONS · TEAM MANAGER

Warsaw Open Data Hackathon 2026 · Top 10, Challenge 1 (Warsaw 2076) — Oct 2026

- Managed a team of 5 through GitHub issues and reviewed pull requests over 24 hours, and owned the architecture, API contract and integration of Find YOUR way, a guided walk through Warsaw on open data
- Built the FastAPI backend (104 pytest tests): OSRM street routing with an offline route cache, an LLM that picks an outing from the user's words, a SQLite store of city layers and an MCP server for AI agents

HackYeah 2026 · Top 10, Artificial Intelligence track — Oct 2026

- Led a team of 3 and built the backend of GdzieZamieszkać, which compares Kraków's 18 districts on 51 public-data measures: a FastAPI data API with 16 operations and 161 offline pytest tests on PostgreSQL + PostGIS
- Added an LLM report service (OpenRouter) with a guard that drops any generated text containing a number not in the source data; deployed with Docker, Caddy and Cloudflare Tunnel

Codaro Coding Challenge II · 1st in Track B (booking and scheduling), 2nd overall — Aug 2026

- Managed a team of 5 and co-developed the FastAPI backend of Arbor, a config-driven booking engine where a new business niche needs one JSON file instead of a rewrite (Next.js 14, Supabase PostgreSQL with row-level security, JWT verification via JWKS, Docker)

42 Warsaw Hacks · 3rd place — Aug 2026

- Built ft_WarsawStories with 1 teammate: a live campus dashboard over the 42 Intra API, with a cache-first FastAPI + SQLite backend, scheduled sync jobs, stale-cache fallback and a React 19 + TypeScript frontend

RESEARCH EXPERIENCE

Bioinformatician, Doctoral Researcher · Nencki Institute of Experimental Biology, Warsaw — Sept 2021 – Dec 2025

- Built Python, R and Bash pipelines and in-house databases to process multi-terabyte sequencing data from 3 primate species across 6 assay types
- Wrote a pipeline that separates mixed-species sequencing reads from co-cultured cells
- Co-authored 3 papers in *Cell Stem Cell*, *Nature Cell Biology* and *Nature Communications*; Director's Award for Distinguished Publication 2024
- Supervised 4 interns; Student Council Secretary, 2023 – 2025

Doctoral Intern · Max Planck Institute for Molecular Genetics, Berlin — Oct 2024 – Nov 2024
Junior Research Fellow · JNCASR, Bengaluru — Aug 2020 – July 2021

EDUCATION

Software Engineering, Core Curriculum · 42 Warsaw — June 2026 – Present
Peer-to-peer, project-based programme in C and systems programming. Admitted through the 4-week Piscine (C and Shell, Feb 2026).

M.Sc. Bioinformatics and Biotechnology · IBAB, Bengaluru — June 2020
B.Tech. Biotechnology · Neotia Institute of Technology, Management and Science, Kolkata — July 2018

CERTIFICATES AND LANGUAGES

Introduction to Machine Learning, Duke University (Coursera), May 2021

English (C2, EF SET 75/100) · Bengali, Hindi (native) · Polish (B1, limited working) · French, German (elementary)